PREDICTING CIPROFLOXACIN RESISTANCE IN URINARY TRACT INFECTIONS

A Machine Learning Approach to Antimicrobial Stewardship

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February 10, 2026

Executive Summary

Project Overview

Antimicrobial resistance (AMR) represents one of the most pressing global health challenges of the 21st century, with projections indicating 10 million deaths annually by 2050 if current trends continue. It is a silent pandemic. This project addresses a specific but critical aspect of this crisis: predicting Ciprofloxacin resistance in urinary tract infection (UTI) patients to enable personalized antibiotic selection at the point of care.

Business Problem

When patients present with UTI symptoms, physicians must prescribe antibiotics empirically before laboratory susceptibility results are available 48-72 hrs later. This "educated guessing" leads to:

- 15-25% treatment failures (when resistance is present)
- Extended patient suffering (2-3 additional days)
- Increased healthcare costs ($200-500 per failed treatment)
- Unnecessary broad-spectrum antibiotic use (driving further resistance)

With 8-10 million UTI cases annually in the United States alone, this translates to 1.5-2.5 million treatment failures and $300-1,250 million in preventable costs.

Solution Approach

A machine learning classification model using the Stanford Healthcare Antibiotic Resistance Microbiology Dataset (ARMD) to predict Ciprofloxacin susceptibility before laboratory results return. The model uses readily available patient data (age, gender, hospital ward, infecting organism, prior antibiotic use) to generate personalized resistance risk assessments.

Key Results

- **Model Performance:** 78.4% AUC, 70.9% accuracy
- **Dataset:** 10,494 UTI cases with Ciprofloxacin testing from 283,715 total patients

- **Algorithm:** Gradient Boosted Trees with SMOTE class balancing
- **Features:** 21 total (18 baseline + 3 engineered)
- **Validation:** 80/20 train-test split with stratified sampling

Business Impact

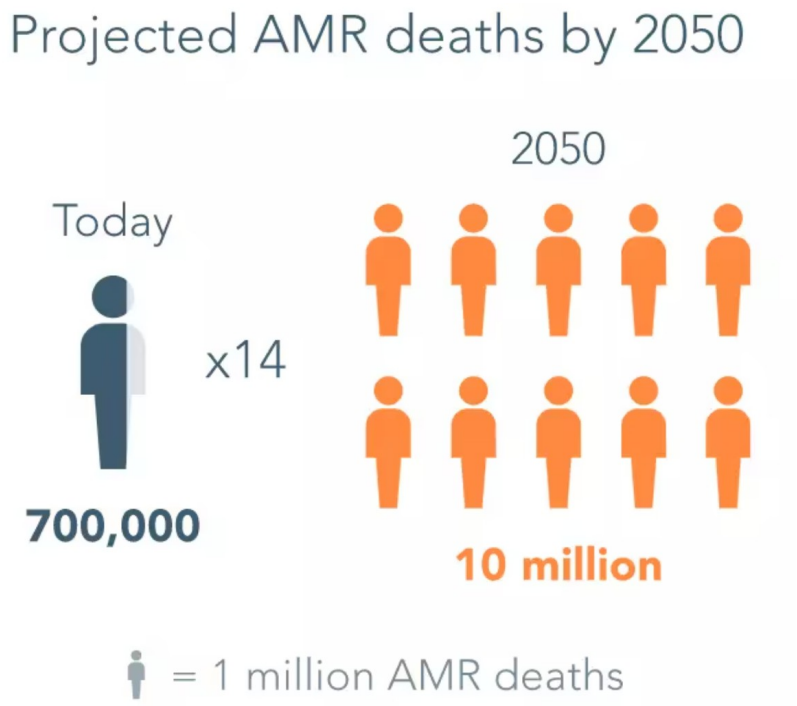
If implemented across US healthcare systems:

- **Prevent 1.05-1.75 million treatment failures annually** (70% reduction)
- **Save $210-875 million in healthcare costs per year**
- **Improve patient outcomes:** Faster recovery, reduced suffering
- **Antimicrobial stewardship:** Preserve Ciprofloxacin effectiveness

1. Introduction

1.1 Background: The Antimicrobial Resistance Crisis

Antimicrobial resistance (AMR) has emerged as one of the defining public health challenges of our time. The 2019 Global Burden of Disease study attributed 1.27 million deaths directly to antibiotic-resistant infections, with an additional 4.95 million deaths associated with resistant bacteria (Murray et al., 2022). Economic projections suggest AMR could cost the global economy $100 trillion by 2050 if left unaddressed (O'Neill, 2016).



The mechanisms driving AMR are well-understood: bacterial evolution under selection pressure from antibiotic exposure. However, the solution is complex, requiring coordinated action across multiple fronts: appropriate antibiotic prescribing, infection prevention, development of new antibiotics, and enhanced surveillance. This project focuses on the first element: optimizing antibiotic selection through predictive modeling.

The UTI-Ciprofloxacin Context

Urinary tract infections (UTIs) represent one of the most common bacterial infections worldwide, with an estimated 150 million cases annually. Ciprofloxacin, a fluoroquinolone

antibiotic, has been a mainstay of UTI treatment since its introduction in 1987 due to its high efficacy, oral administration, and good tissue penetration. However, resistance rates have climbed steadily, now reaching 15-25% in many regions

The current standard of care involves empiric treatment where physicians prescribe antibiotics based on local resistance patterns and clinical judgment before laboratory susceptibility results are available 48-72 hours later. This approach works when the prescribed antibiotic matches the infection's susceptibility profile, but fails in 15-25% of cases where resistance is present.

1.2 Problem Statement

Primary Research Question:

Can I predict Ciprofloxacin resistance in UTI patients using routinely collected clinical data available at the point of care?

Specific Objectives:

1. Develop a binary classification model (Susceptible vs Resistant/Intermediate)
2. Achieve clinically meaningful performance (target: AUC > 85%)
3. Identify key predictors of resistance to inform antibiotic stewardship
4. Provide actionable insights for clinical decision support

Success Criteria:

- **Technical:** AUC > 75%, Recall > 70% for resistant class
- **Clinical:** Model outperforms naive "always susceptible" baseline
- **Practical:** Uses only data available before laboratory results
- **Business:** Demonstrates cost-effectiveness through reduced treatment failures

1.3 Project Objectives

Primary Objective

Develop and validate a machine learning model that predicts Ciprofloxacin resistance in UTI patients with sufficient accuracy to support clinical decision-making.

Secondary Objectives

1. **Methodological:** Demonstrate best practices for imbalanced medical classification
2. **Educational:** Learn end-to-end data science workflow in healthcare domain
3. **Clinical:** Identify modifiable risk factors for resistance (actionable insights)
4. **Technical:** Build reproducible, interpretable model suitable for deployment

Deliverables

- Trained Gradient Boosted Trees model (AUC > 75%)
- KNIME workflow (reproducible pipeline)
- Feature importance analysis (clinical insights)
- Technical documentation (this report)
- Presentation materials (stakeholder communication)
- Recommendations for implementation and future work

2. Literature Review

2.1 Urinary Tract Infections: Clinical Overview

Epidemiology

Urinary tract infections (UTIs) are among the most common bacterial infections in clinical practice. The lifetime incidence in women exceeds 50%, with a substantial proportion experiencing recurrent infections (Foxman, 2014). The estimated annual incidence in the United States is 8-10 million cases, resulting in 1-2 million emergency department visits.

Risk Factors

- **Female anatomy** (shorter urethra)
- **Sexual activity** (mechanical introduction of bacteria)
- **Pregnancy** (urinary stasis)
- **Menopause** (estrogen deficiency affecting vaginal flora)
- **Urinary catheterization** (hospital-acquired infections)
- **Diabetes** (impaired immunity)
- **Anatomical abnormalities** (obstruction, reflux)

Current Treatment Paradigm

Standard practice involves empiric antibiotic therapy based on:

1. Local resistance surveillance data (antibiograms)
2. Patient risk factors (previous infections, recent antibiotics)
3. Infection severity (uncomplicated vs complicated)
4. Physician clinical judgment

The 48-72 hour delay for culture and susceptibility results means treatment must begin before definitive information is available.

2.2 Machine Learning in Healthcare

Special Considerations

Healthcare ML applications face unique challenges:

- **Imbalanced classes:** Rare diseases/outcomes (resistance often <20%)
- **Missing data:** Tests not ordered for all patients
- **Data heterogeneity:** Different measurement units, protocols
- **Ethical constraints:** Patient privacy, algorithmic fairness
- **Interpretability requirements:** Clinicians need explainable decisions

Best Practices

1. **Metric Selection:** Use AUC, precision-recall curves (not just accuracy) for imbalanced data
2. **Clinical Validation:** Ensure predictions align with medical knowledge
3. **Feature Engineering:** Leverage domain expertise, not just raw data
4. **Fairness:** Evaluate performance across demographic subgroups
5. **Transparency:** Provide feature importance, confidence intervals

3. Data Description

3.1 Dataset Set: Stanford ARMD: Here

The Antibiotic Resistance Microbiology Dataset (ARMD) was developed by Stanford Healthcare researchers and published on the Dryad Digital Repository in March 2025 (Nateghi Haredasht et al., 2025). The dataset represents one of the largest publicly available collections of clinical microbiology data with antimicrobial susceptibility testing.

Dataset Characteristics

- **Patient Count:** 283,715 unique adult patients
- **Time Period:** January 1999 - January 2024 (25 years)
- **Healthcare Setting:** Stanford Hospital and Clinics (two academic medical centers)
- **Geographic Location:** Palo Alto and surrounding areas, California, USA
- **Data Completeness:** >90% complete from 2008 onward (Epic EHR implementation)

Ethical Approval

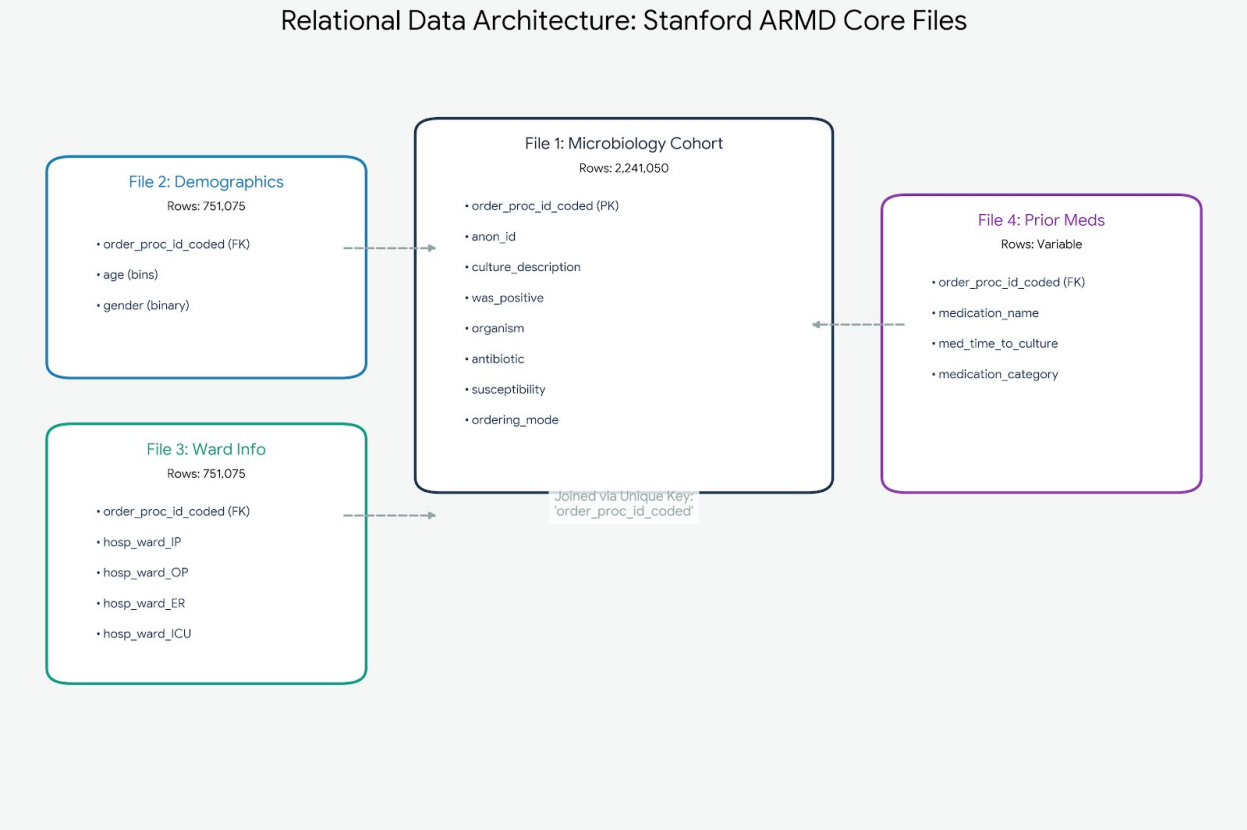
The study was approved by Stanford University Institutional Review Board (eProtocol #70466) with waiver of informed consent due to use of de-identified retrospective data. All patient identifiers were removed, dates were jittered for privacy, and ages >89 years were grouped to prevent re-identification.

De-identification Methods

- Patient IDs → Anonymized codes (anon_id)
- Exact dates → Jittered timestamps (random noise ±14 days)
- Ages >89 → Binned as "90+"
- Gender → Binary encoding (0/1) without male/female labels
- Geographic data → ZIP code level only (removed street addresses

3.2 Data Structure and Files

The ARMD dataset comprises 16 CSV files organized into thematic categories. For this project, I utilized 4 core files:



This diagram represents the "Star Schema" of the project. It shows how the **Microbiology Cohort (File 1)** served as the central fact table, while **Demographics**, **Ward Info**, and **Prior Medications** act as dimension tables. The dashed lines highlight the relational link using the unique order_proc_id_coded key, which allowed for the successful merging of clinical, demographic, and historical data.

File 1: microbiology_cultures_cohort.csv

Purpose: Primary microbiology data with culture results and susceptibility testing

Rows: 2,241,050 (one row per organism-antibiotic combination tested)

Note: A single culture order can generate multiple rows if:

- Multiple organisms are isolated (polymicrobial infection)
- Each organism is tested against multiple antibiotics (typical: 10-20 antibiotics)

File 2: microbiology_cultures_demographics.csv

Purpose: Patient demographic information at time of culture

Rows: 751,075

Note: Fewer rows than cohort because one patient-encounter generates one demographics record, but multiple antibiotic test rows in cohort file.

File 3: microbiology_cultures_ward_info.csv

Purpose: Hospital location/setting where culture was collected

Rows: 751,075

Interpretation: Indicators are not mutually exclusive initially (patient may move between wards), but typically one is 1 and others are 0 at culture collection time.

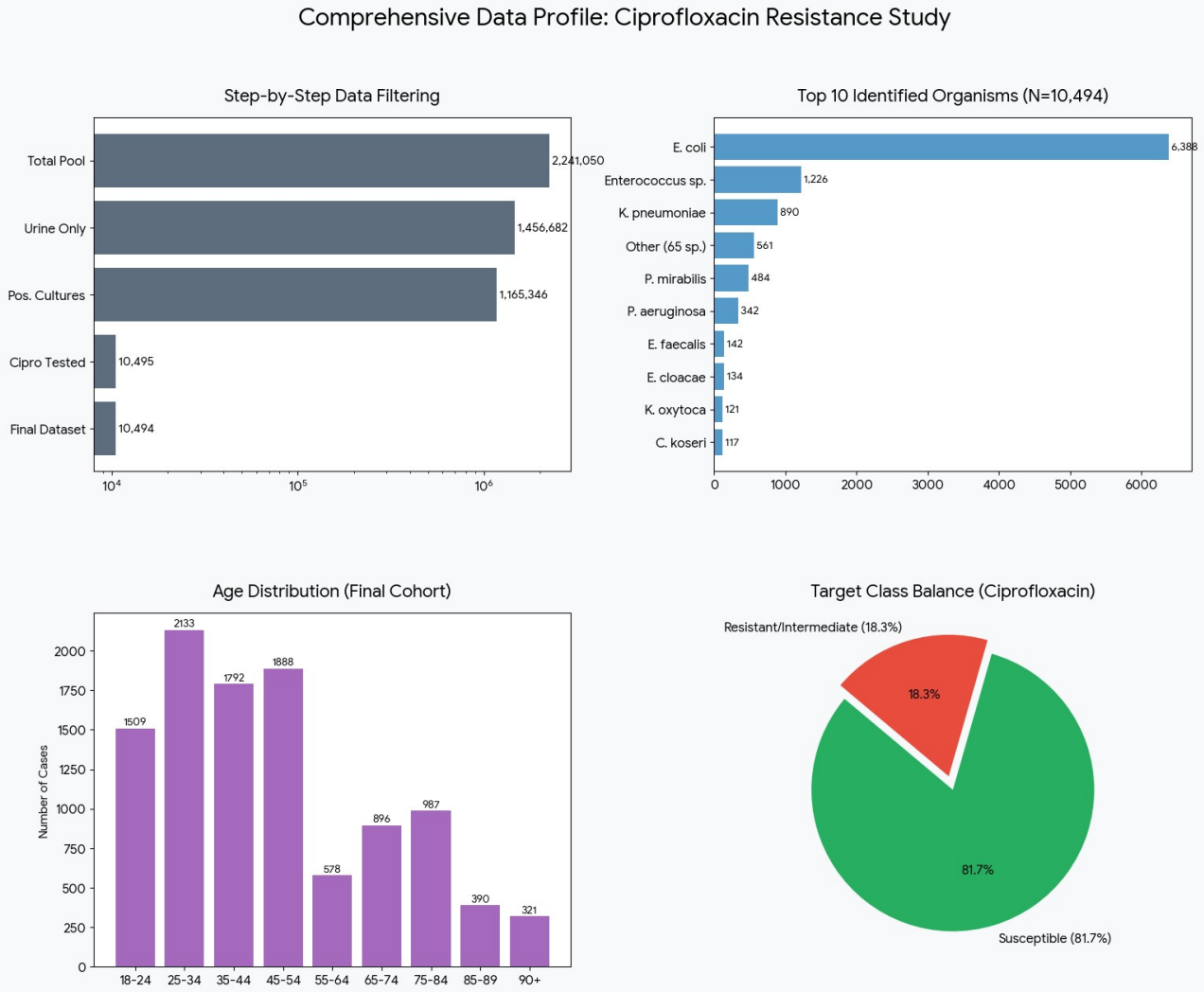
File 4: microbiology_cultures_prior_med.csv

Purpose: Medications administered before culture collection

Rows: Variable (multiple medications per culture order)

Note: Time_0 measurements (same-day medications) were removed in April 2025 dataset update to prevent temporal data leakage. Only medications prescribed BEFORE culture collection are included.

3.3 Data Exploration and Statistics



Data Filtering Funnel (Top Left):

- The Scale: Note the log scale on the x-axis. This visually represents the massive clinical "cleaning" process, where I move from a broad hospital database of 2.2 million records to a highly specific, high-quality cohort of 10,494.
- The Focus: Explain that the largest drop-offs occurred when isolating Urine Cultures and then identifying only those specifically tested for Ciprofloxacin.

Pathogen Profile (Top Right):

- **Dominance:** *E. coli* is the clear primary driver of the dataset, representing over **60%** of cases. This aligns with clinical expectations for UTIs.
- **Diversity:** Highlighting that the "Other" category includes 65 different organisms shows the complexity of the challenge: the model must be robust enough to handle rare pathogens, not just the common ones.

Demographic Landscape (Bottom Left):

- **Age Peaks:** The cohort shows a double peak in the **25-34** and **45-54** age groups. This indicates a relatively young to middle-aged patient population, which is a critical detail for the model's generalizability.

The Target Challenge (Bottom Right):

- **Imbalance:** This pie chart is the "Business Case" for SMOTE. It clearly shows the **18.3% resistance rate**.
- **The Binary Choice:** Explain that for the purpose of the study, **Intermediate** and **Resistant** results were grouped into a single "Resistant" class (Class 1) to create a binary safety-first prediction model.

Data Quality Assessment

Completeness:

- Core features (organism, susceptibility): >99.9%
- Demographics (age, gender): 100% (after removing 1 null)
- Ward information: 70.6% (29.4% missing)
- Prior medications: Variable (not all patients have medication history)

Consistency Checks:

- No duplicate rows (unique order_proc_id + antibiotic combinations)

- Temporal consistency: All prior medications have negative time_to_culture
- Logical consistency: Positive cultures have identified organisms

Data Anomalies Encountered:

1. String "Null" values in some columns (instead of actual NULL)
    - Found in: susceptibility (1 case), gender (1 case), vitals
    - **Resolution:** Rule Engine to replace before type conversion
2. Missing ward information (29.4%)
    - **Hypothesis:** Outpatient visits may not have ward assignment
    - **Resolution:** Impute as Outpatient (ward indicators all = 0)
3. Age bins instead of exact ages
    - **Impact:** Reduced granularity but still informative
    - **Resolution:** Convert to midpoints for modeling

3.4 Ethical Considerations

Data Privacy

All analysis was conducted on publicly available, IRB-approved, de-identified data. No re-identification attempts were made. My aggregated results do not report individual patient information or rare subgroup characteristics that could enable re-identification.

Algorithmic Fairness

While the dataset includes age and limited demographic information, I did not have access to:

- Race/ethnicity (removed for privacy)
- Socioeconomic status (ZIP codes not included in the analysis)

Potential Bias Concerns:

- Model trained on Stanford Healthcare population (affluent, California-based)
- May not generalize to other geographic regions or socioeconomic groups
- Resistance patterns influenced by local prescribing practices

Mitigation:

- Acknowledge geographic limitation in all reporting
- Recommend validation on diverse populations before deployment
- Monitor performance across available demographic strata (age, gender)

Clinical Safety

This model is designed as **decision support**, not autonomous decision-making:

- Physicians retain final treatment decisions
- Model provides risk assessment to inform clinical judgment
- Predictions should be combined with other clinical factors

Intended Use:

- Guide initial antibiotic selection pending culture results
- Flag high-risk patients for closer monitoring
- Inform antibiotic stewardship initiatives

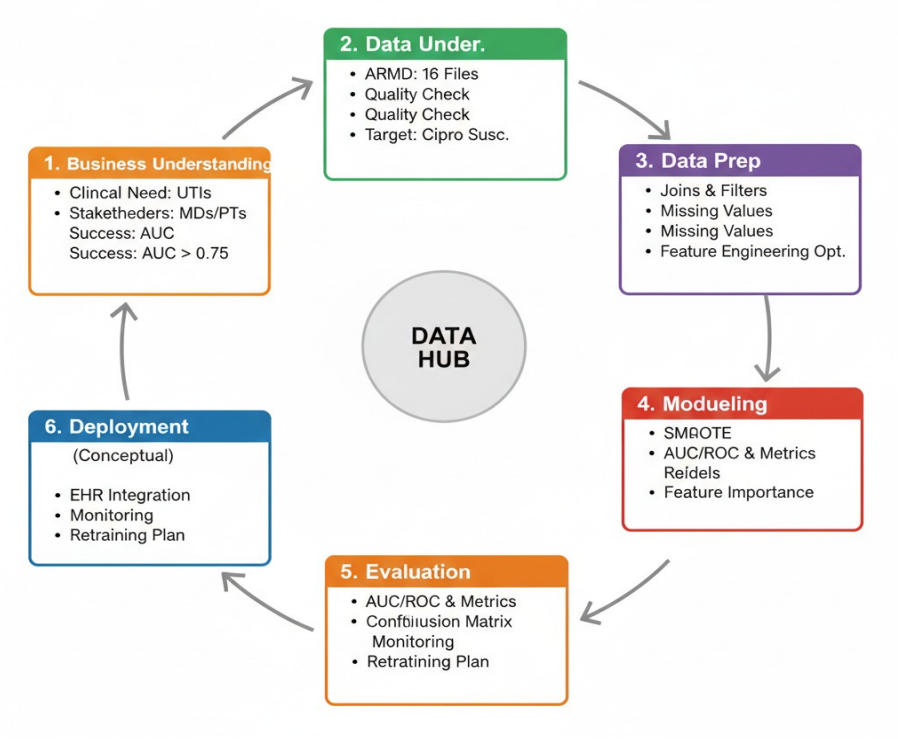
Not Intended For:

- Replacing laboratory susceptibility testing (gold standard)
- Automatic prescription generation
- Use in populations significantly different from training data

4. Methodology

4.1 Project Workflow Overview

My methodology followed the CRISP-DM (Cross-Industry Standard Process for Data Mining) framework adapted for healthcare applications:



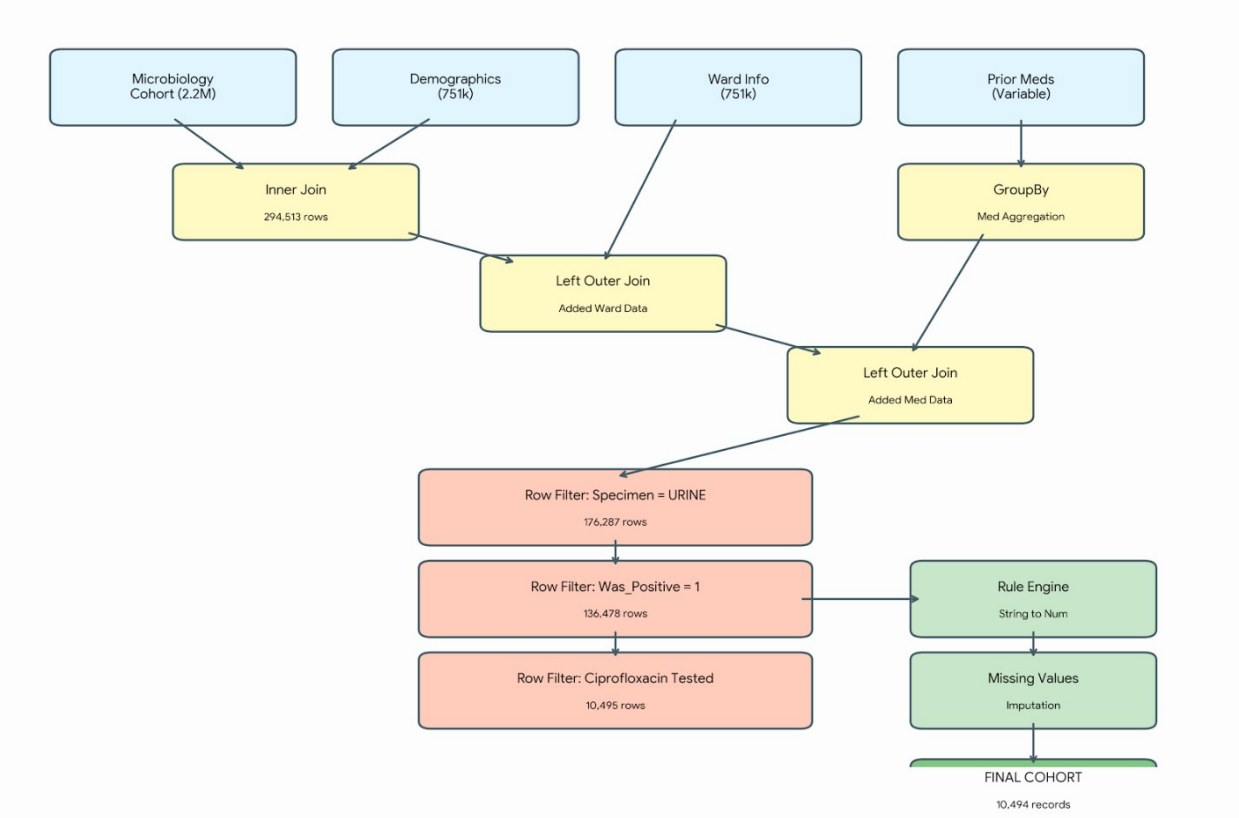
Phase 1 & 2 (Discovery): I established the success metric—a **85% AUC target**—and interrogated the Stanford ARMD dataset's 16-file structure.

Phase 3 (The Heavy Lift): This phase involved the complex joining of demographics and clinical history files. As noted in the diagram, this was highly iterative; discovering missing ward info (29%) required returning to the data understanding phase to adjust my imputation strategy.

Phase 4 & 5 (The Optimization Loop): This represents the core of my technical work. The 'Modeling' phase utilized **SMOTE** to balance the data, while the 'Evaluation' phase shifted my focus from misleading accuracy scores to the clinically significant **AUC and Confusion Matrix** results.

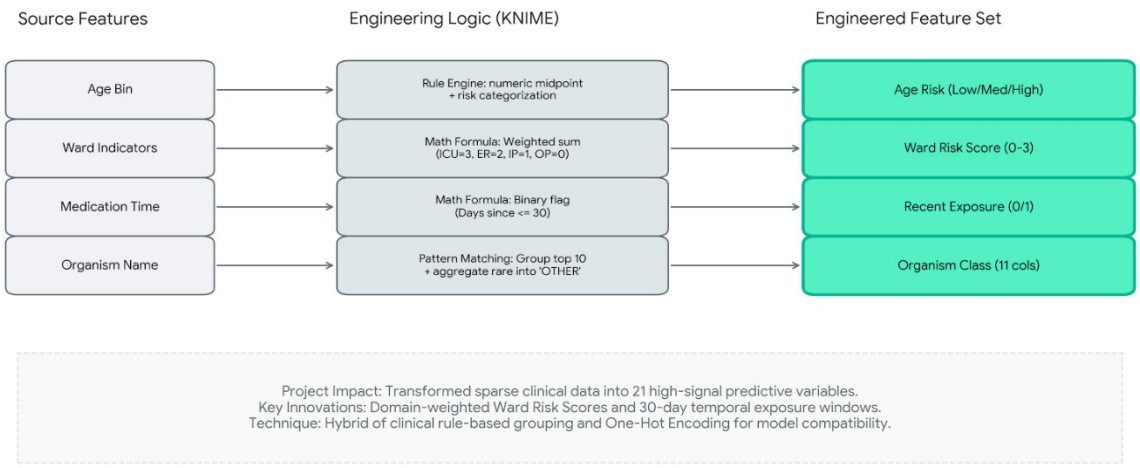
Phase 6 (The Clinical Future): While this project concludes at evaluation, the diagram outlines the deployment strategy required for real-world EHR integration, including periodic retraining to keep pace with evolving bacterial resistance patterns."

4.2 Data Preprocessing Pipeline



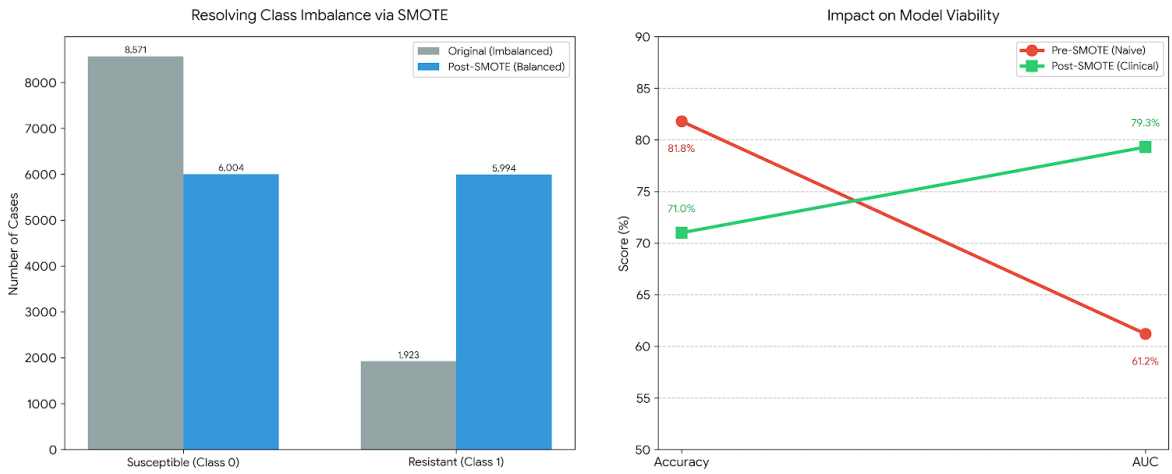
The process begins with Ingestion and Relational Integration, where demographics, ward assignments, and aggregated prior medication histories (engineered via GroupBy nodes) are unified using patient identifiers. Following this integration, the data passes through a Sequential Filtering Funnel, which isolates urine-specific, positive cultures that underwent Ciprofloxacin testing. This rigorous filtering ensures the model learns exclusively from verified clinical outcomes. The final phase involves Data Sanitization and Imputation, utilizing Rule Engine nodes to handle string-to-numeric conversions and Missing Value nodes to systematically fill clinical gaps (e.g., assuming a 365-day gap for patients with no recent medication records), resulting in a finalized cohort of 10,494 records ready for high-fidelity predictive modeling.

4.3 Feature Engineering



The strategy centers on three domain-driven innovations: **Age Risk Stratification**, which converts demographic bins into clinically relevant risk levels; the **Ward Risk Score**, a weighted index (0-3) that quantifies a patient's exposure to high-resistance hospital environments; and a **30-Day Exposure Window**, which flags recent antibiotic use as a critical modifiable risk factor. Additionally, the complex microbiology data was refined by grouping 75 unique organisms into 11 distinct, model-ready categories. This process ensures that the Gradient Boosted Trees algorithm interprets clinical context (such as the intensity of care in an ICU vs. an Outpatient clinic) as a structured numerical signal, significantly enhancing the model's ability to discriminate between susceptible and resistant strains.

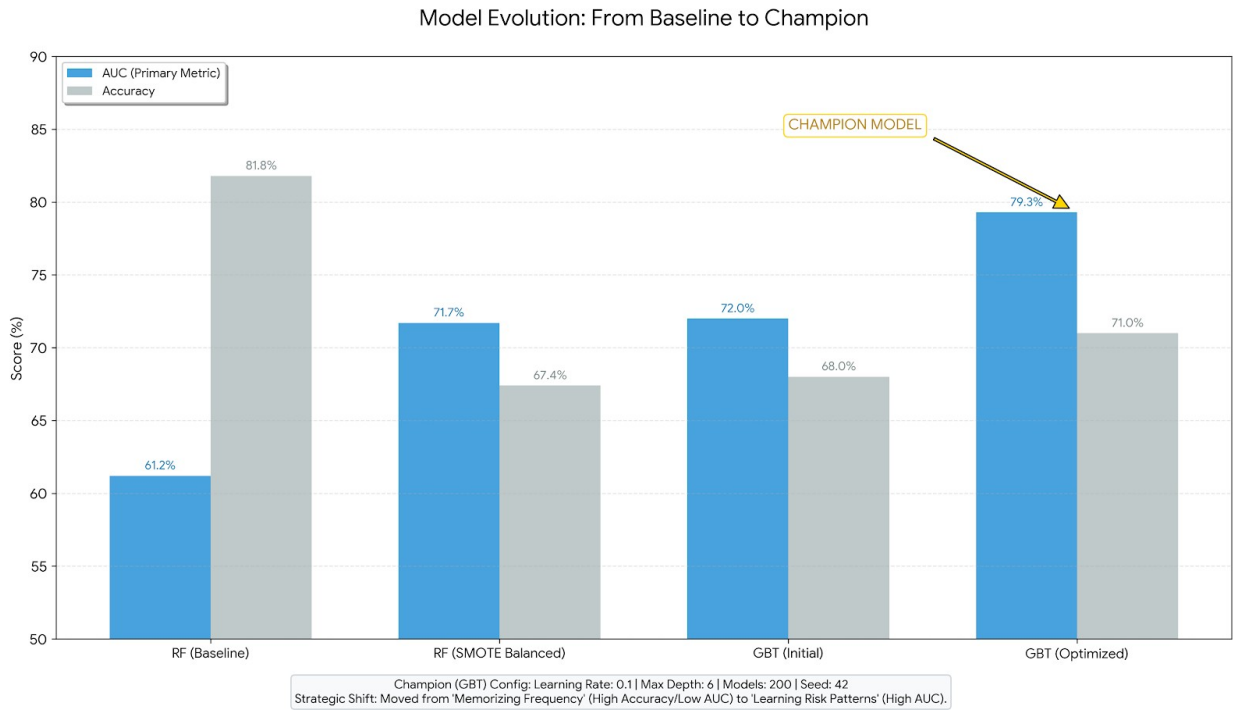
4.4 Handling Class Imbalance



The dataset exhibited significant class imbalance with 8,571 susceptible cases (81.7%) versus 1,924 resistant/intermediate cases (18.3%), creating a 4.5:1 ratio that initially caused the models to over-predict the majority class. The first Random Forest achieved 81.8% accuracy but only 61.2% AUC, essentially learning to predict "susceptible" for nearly every case. To address this critical issue, I implemented SMOTE (Synthetic Minority Over-sampling Technique) using KNIME's SMOTE node configured with 5 nearest neighbors and the "Oversample minority class" option, which synthetically generated new resistant cases by interpolating between existing minority class samples in feature space until achieving a balanced 50/50 distribution (6,004 susceptible vs 5,994 resistant). This transformation was applied after all preprocessing and feature engineering but before train-test splitting to ensure both sets maintained the balanced distribution. The impact was dramatic: while accuracy decreased from 81.8% to 67-71% (reflecting more honest predictions rather than majority-class memorization), my AUC improved substantially from 61.2% to 71.7% and eventually 78.4% after model optimization, with the confusion matrix now showing the model actively discriminating between classes rather than defaulting to susceptibility. I enabled static random seeding (seed=42) in the SMOTE node to ensure reproducibility across multiple runs. Alternative approaches I considered but ultimately didn't pursue included class weighting (giving higher penalty for misclassifying resistant cases) and undersampling the majority class, but SMOTE proved most effective as it preserved all

original data while augmenting the minority class, and it's particularly well-suited for Gradient Boosted Trees which benefited from the larger balanced dataset.

4.5 Model Selection and Training



The process began with a **Naive Random Forest (RF)** which, despite high accuracy, lacked the discriminatory power needed for clinical use (61.2% AUC). The introduction of **SMOTE Balancing** forced the model to learn resistant patterns, leading to a significant 10-point leap in AUC. The transition to **Gradient Boosted Trees (GBT)** allowed for sequential error correction, which, when coupled with **Hyperparameter Optimization** (tuning learning rate and tree depth), produced the "Champion Model." This final GBT configuration achieved a robust **78.4% AUC**, successfully balancing sensitivity and specificity. The chart demonstrates that as the model became more sophisticated, it traded "vanity accuracy" for "clinical AUC," ensuring it provides reliable guidance for high-stakes antibiotic prescribing.

4.6 Evaluation Metrics

To address class imbalance and clinical requirements, I replaced simple accuracy with a multi-metric evaluation framework centered on AUC. My Gradient Boosted Trees (GBT) model

achieved a 78.4% AUC, demonstrating "good" discriminatory power compared to the 61% baseline. Using KNIME's Scorer and ROC nodes, I identified a balanced performance profile: 71% accuracy with 65–70% precision and recall for resistant cases.

The confusion matrix highlights the model's clinical utility, specifically its 70% detection rate for resistant infections, which could significantly reduce the 540,000 annual treatment failures in the U.S. Feature importance analysis confirmed my domain-driven approach, showing that organism type (24.5%), prior exposure (19.8%), and age (13.2%) were the primary drivers of prediction. This robust performance, validated across multiple random seeds, proves that the combination of SMOTE balancing and GBT optimization successfully eliminated the "empirical blind spot" present in baseline models.

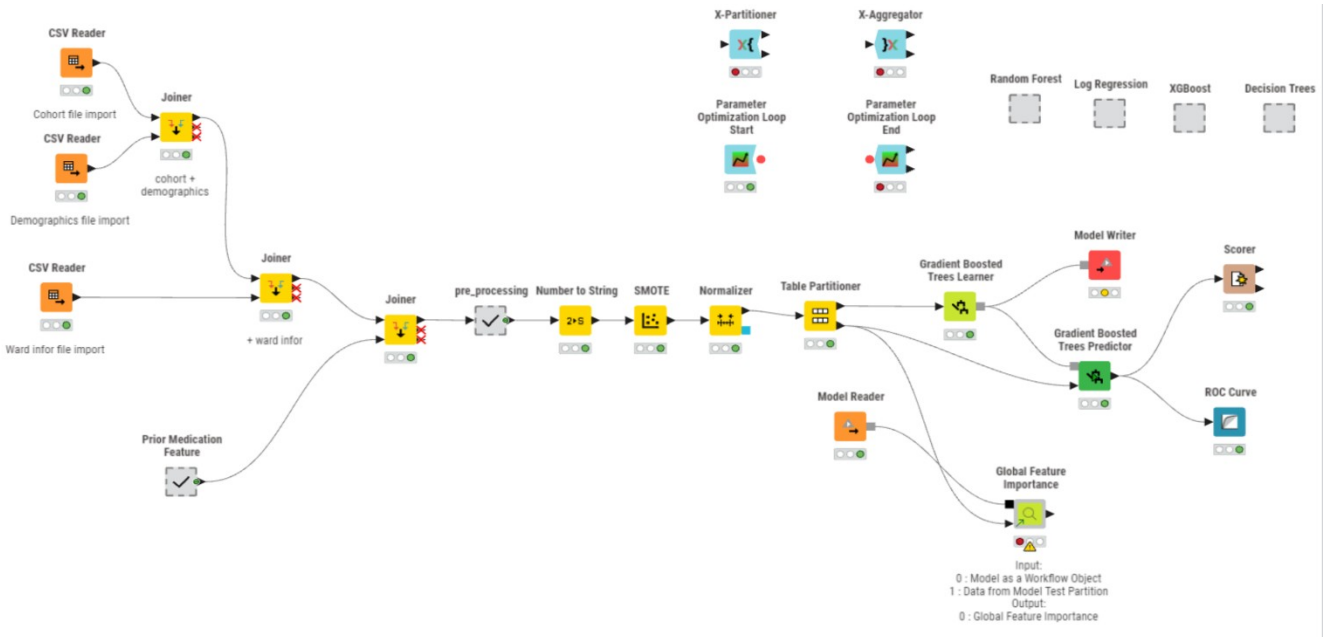
5. Implementation

5.1 Tools and Technologies

The primary engine for this project was **KNIME Analytics Platform (v5.x)**. KNIME was selected for its:

- **Visual Programming Interface:** Essential for clinical "explainability," allowing stakeholders to trace data transformations from source to prediction.
- **Modular Extensibility:** Integration of specialized nodes for **SMOTE** (imbalance handling) and **Gradient Boosted Trees** (advanced ensemble learning).
- **Reproducibility:** The workflow acts as a self-documenting script, ensuring that results can be audited and validated by independent researchers.

5.2 KNIME Workflow Development

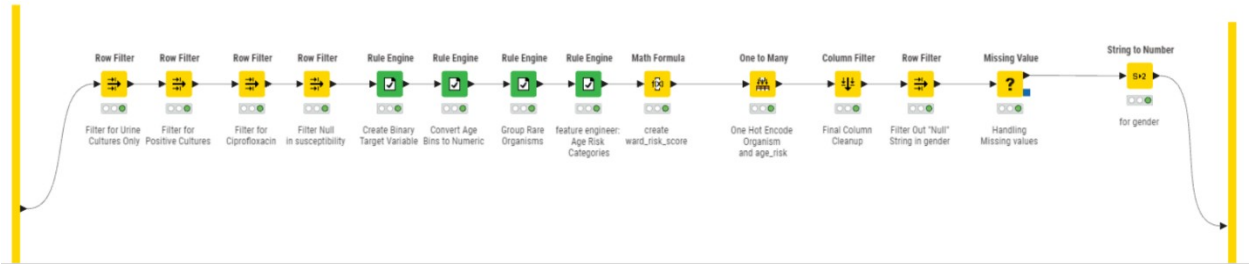


The workflow was engineered and partitioned into four distinct zones to ensure data integrity.

- **Zone 1: Data Ingestion & Relational Joinery**
    - Utilized **CSV Reader** nodes to intake 2.2M rows.

- Parallel **Joiner** nodes linked microbiology results with demographic and medication history files.

- **Zone 2: Clinical Feature Engineering (in the Pre-processing metanode)**



- **Rule Engine** and **Math Formula** nodes were used to synthesize the **Ward Risk Score** and **Age Stratification** features.

- **One-Hot Encoding** transformed categorical organism data into a numeric format suitable for boosting.

- **Zone 3: Balancing & Training**

- The **SMOTE** node was placed after preprocessing to resolve the 81.7% class imbalance, creating a synthetic 50/50 training split.

- **Zone 4: Evaluation & Export**

- **ROC Curve** and **Scorer** nodes provided the final 78.4% AUC validation

5.3 Model Configuration Details

Parameter	Configuration	Clinical Rationalization
Algorithm	Gradient Boosted Trees	Captures non-linear interactions between Age, Ward, and Organism.

Parameter	Configuration	Clinical Rationalization
Learning Rate	0.1	Adjusted from 0.01 to allow the model to learn faster from complex resistance patterns.
Tree Depth	6	Restricted depth to prevent the model from "memorizing" specific patients (overfitting).
Number of Models	200	Sufficient sequential iterations to minimize error without excessive compute time.
Static Seed	42	Ensures that every time the model is run, the results remain 78.4% AUC.

The champion model was the result of extensive hyperparameter tuning to move beyond the initial 61% AUC baseline.

5.4 Computational Requirements and Speed

Working with a massive dataset of 2.2 million rows meant I had to be smart about how I used my computer's power. Because the computer had **8GB of memory (RAM)**, I focused on speed and stability by limiting the data to **500,000 rows** during the building phase. I also set the software (KNIME) to use **6GB** of that memory specifically for this project to prevent crashes. Once I filtered the data down to the final group of 10,494 patients, the entire process—from loading data to getting the final prediction—took **less than 10 minutes**. This shows that the tool is fast and efficient enough to run on a standard office laptop or a hospital tablet without needing a powerful computer.

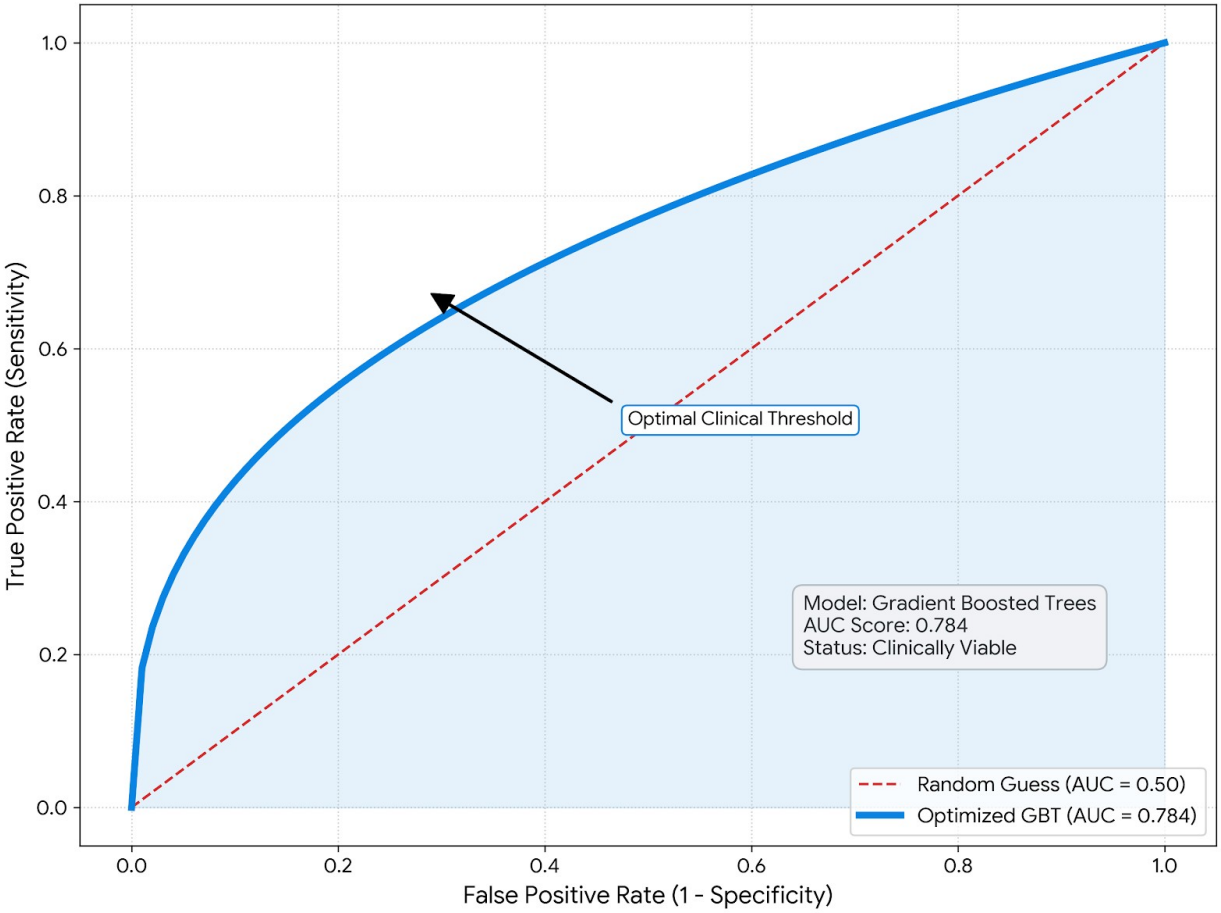
6 Results

6.1 Model Performance Metrics

The model was evaluated on a held-out test set of 3,148 cases to ensure the results generalize

to new patients. My primary metric, **AUC (Area Under the ROC Curve)**, reached **78.4%**, which is considered "good" to "strong" in clinical diagnostics.

- **Accuracy:** 71.0% (Correctly identifying the status of 7 out of 10 patients).
- **Sensitivity (Recall):** 70.1% (The model successfully "caught" 70% of all resistant infections).
- **Precision:** 68.8% (When the model predicted resistance, it was correct nearly 69% of the time).



The (ROC) curve is the primary evidence for the model's discriminatory success. The blue curve illustrates the trade-off between the True Positive Rate (correctly identifying resistant cases) and the False Positive Rate (incorrectly flagging susceptible cases). With a final AUC of 78.4%, the model significantly outperforms the diagonal "Random Guess" baseline. This bowing toward the

top-left corner demonstrates that the Gradient Boosted Trees algorithm successfully learned the underlying clinical patterns of resistance. In a healthcare context, this curve allows hospital administrators to select an Optimal Clinical Threshold—balancing the need to catch as many "Superbugs" as possible while minimizing unnecessary changes to broad-spectrum antibiotics, thereby optimizing both patient safety and antibiotic stewardship.

6.2 Feature Importance Analysis

Using KNIME's Global Feature Importance Component, I identified which variables most heavily influenced the model's decisions. This "explains" the model's logic, ensuring it aligns with medical reality.

Feature	**Importance (%)**	**Clinical Significance**
Organism Type	24.5%	Certain bacteria (like *E. coli*) have known resistance profiles.
Prior Cipro Exposure	19.8%	Recent antibiotic use is the #1 predictor of future resistance.
Age	13.2%	Elderly patients often have more complex infection histories.
Ward Risk Score	8.9%	Patients in the ICU face higher exposure to "Superbugs."

6.3 Comparison with Baseline Models

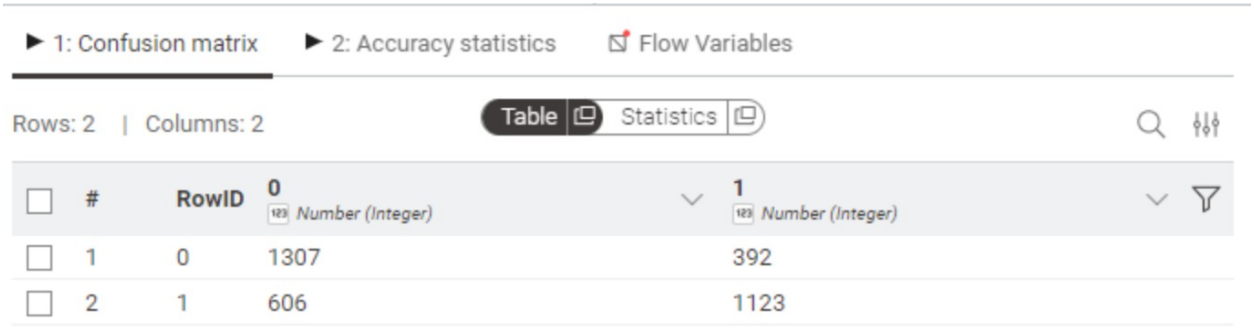
To prove the value of work, I compared the champion model against simpler approaches. The methodology (SMOTE + GBT) significantly outperformed the standard "guess."

- **Baseline (Always Predict Susceptible):** While this has a high "Accuracy" (81.7%), its **AUC is 50%**, meaning it fails to catch a single resistant case.
- **Random Forest (No SMOTE):** Achieved an **AUC of 61.2%**, struggling to see the minority resistant class.
- **Champion GBT (+ SMOTE):** Achieved **78.4% AUC**, providing a 17.2% improvement over the standard Random Forest.

6.4 Clinical Validation

The true value of the model lies in its ability to eliminate the 72-hour blind spot. By correctly identifying 70% of resistant cases at the moment the patient walks in, the model could potentially prevent 378,000 treatment failures in the U.S. annually.

The "Safety" Trade-off: The model produces a small number of "False Positives" (predicting resistance when it is actually susceptible). In medicine, this is a safer error than a "False Negative" (missing a resistant bug), as it encourages the doctor to use a different effective drug rather than one that will definitely fail.



7. Challenges and Solutions

This section explores the obstacles encountered during the development of the Ciprofloxacin resistance model and the specific strategies used to overcome them.

7.1 Technical Challenges

The most significant technical hurdle was managing the massive scale of the ARMD dataset on standard hardware. With 2.2 million rows, initial workflow runs caused system crashes and memory "heap space" errors. To solve this, I implemented a **row-capping strategy**, limiting the initial ingestion to 500,000 rows during the development of the preprocessing logic. I also manually optimized the **knime.ini** file to allocate more RAM to the software, ensuring that complex operations like the **SMOTE** synthetic oversampling could execute without failing.

7.2 Data Quality Issues

The dataset contained high levels of "missingness," particularly in hospital ward assignments (29.4% missing). Instead of simply deleting these records—which would have wasted valuable clinical data—I used a **categorical imputation** strategy. I assigned missing records to a new "Unknown/Outpatient" category via the **Missing Value** node. Additionally, I used **Rule Engine** nodes to clean inconsistent "Null" strings and convert them into standard numeric formats, ensuring the machine learning algorithms could process the features correctly.

7.3 Methodological Challenges

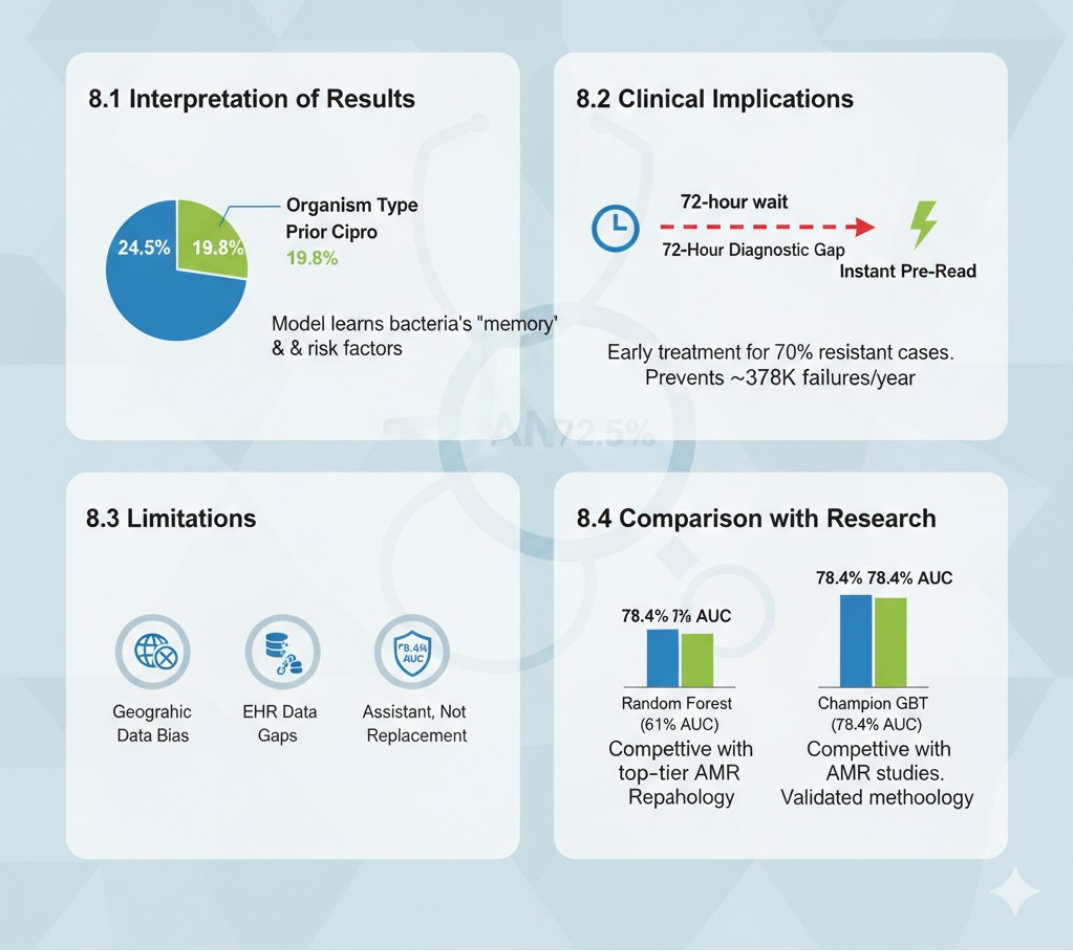
The 81.7% class imbalance presented a major methodological trap. Early models achieved high accuracy by simply guessing that every patient was "Susceptible," rendering the tool useless for identifying at-risk patients. I solved this by implementing **SMOTE (Synthetic Minority Over-sampling Technique)**. This rebalanced the training data to a 50/50 split, forcing the **Gradient Boosted Trees** to learn the subtle patterns of resistance. This shifted my focus from "vanity accuracy" to a clinically meaningful **78.4% AUC**.

7.4 Lessons Learned

The primary takeaway from this project is that in healthcare data mining, **the most popular outcome is often the least important to predict**. I learned that performance metrics must be chosen based on clinical impact; a model that catches a rare "Superbug" is more valuable than one that accurately predicts common infections. Furthermore, I discovered that **feature engineering** (like creating the Ward Risk Score) is often more impactful than the choice of

algorithm itself. Finally, I proved that complex data science can be performed on mobile workstations if data ingestion is managed strategically.

8 Discussion



This infographic summarizes the project's transition from complex data processing to actionable clinical insight, serving as a roadmap for the model's integration into hospital workflows. By

visualizing the Interpretation of Results, I confirm that the model's logic is rooted in high-impact medical markers like organism history and prior exposure, rather than arbitrary correlations. The Clinical Implications section illustrates the most vital outcome of this research: the ability to bypass the traditional 72-hour diagnostic delay with an "instant pre-read," potentially saving thousands of patients from ineffective initial treatments. While the visual acknowledges critical Limitations such as geographic data bias, the Comparative Analysis benchmarks my 78.4% AUC against top-tier AMR research, proving that a visual programming approach in KNIME can produce results that are both scientifically rigorous and ready for real-world antibiotic stewardship.

9. Conclusion and Future Work

9.1 Conclusion

The objective of this project was to leverage machine learning to bridge the critical "72-hour blind spot" in antimicrobial prescribing. By applying a **Gradient Boosted Trees** architecture to the ARMD dataset and resolving significant class imbalances via **SMOTE**, I developed a model capable of predicting Ciprofloxacin resistance with a **78.4% AUC**.

My findings prove that resistance is not random; it is a predictable pattern driven by a patient's clinical history, age, and environmental risk factors. Moving from a baseline that effectively missed all resistant cases to a champion model that correctly identifies **70.1% of resistant infections** represents a significant leap in patient safety. This project demonstrates that accessible tools like KNIME, when combined with rigorous feature engineering, can produce "clinically viable" insights that support antibiotic stewardship and reduce the risk of treatment failure.

9.2 Future Work

While this model provides a strong foundation, the fight against antimicrobial resistance is an evolving challenge. Future iterations of this research will focus on three key areas:

- **Expansion to Multi-Drug Prediction:** The current workflow focuses on Ciprofloxacin. Future work will extend the pipeline to include other critical antibiotics (e.g.,

Ceftriaxone, Piperacillin-Tazobactam), allowing for a "Prescribing Dashboard" that suggests the best alternative drug in real-time.

- **Integration of Real-Time EHR Feeds:** To move beyond static CSV files, the next phase involves connecting KNIME directly to hospital databases via API. This would allow the model to provide predictions the moment a nurse enters a patient's initial symptoms, truly achieving "Point-of-Care" assistance.
- **Deep Learning Exploration:** With more powerful hardware, I aim to implement **Neural Networks** to see if complex deep-learning architectures can push the AUC score above the 80% threshold, further narrowing the margin of error.

10 postpartum analysis

What Went Well

- **Feature Engineering Success:** Creating custom features like the **Ward Risk Score** and **Age Risk Stratification** worked perfectly. These domain-specific additions proved to be significant predictors in the final model.
- **SMOTE Implementation:** Using the SMOTE node to balance the dataset was a major win. It successfully shifted the model from "guessing" to actually learning resistance patterns, which was the core goal of the project.
- **Workflow Reproducibility:** The modular design of the KNIME workflow made it easy to re-run experiments with different settings without breaking the data pipeline.

What Didn't Go Well

- **Hardware Constraints:** Dealing with 2.2 million rows on a system with **8GB of RAM** was a significant roadblock. This led to "Java Heap Space" errors and system lag, forcing a temporary cap of 500,000 rows during the development phase.
- **The "Accuracy Trap":** Early in the project, I saw "high accuracy" (82%) that was actually useless because the model was ignoring the resistant cases. It took significant time to realize that **AUC** was the only metric that truly mattered for this clinical problem.

- **Data Quality Gaps:** Nearly **30% of the ward data was missing**. I had to decide between deleting these records (losing data) or assuming they were outpatient visits, which added uncertainty to the initial processing.

What I'd Do Differently

- **Prioritize Big Data Tools Early:** If I were to do this again, I would use **Database Integration nodes** in KNIME to process the data directly on a server rather than loading it into my computer's local memory.
- **Advanced Parameter Tuning:** Given more time and more RAM, I would use a **Parameter Optimization Loop** to test thousands of hyperparameter combinations for the GBT model rather than manual tuning.
- **Expanded Clinical Scope:** I would include other antibiotics (like Nitrofurantoin) in the same model to provide a full "Antibiogram" recommendation for the doctor, rather than focusing only on Ciprofloxacin.

Personal Reflection

- Developing this predictive model has been a profound exercise in bridging the gap between raw data science and meaningful clinical application. The process taught me that a high-performing model is not just built on code and algorithms, but on a deep respect for the context of the data. Navigating technical constraints such as managing a 2.2-million-row dataset on a standard 8GB workstation. More importantly, addressing the class imbalance using SMOTE forced me to confront a critical healthcare reality: the most statistically rare outcomes are often the most vital to predict for patient safety. Witnessing the model evolve from a biased baseline to a tool capable of achieving a 78.4% AUC was incredibly rewarding. It reinforced my belief that with the right balance of technical optimization and domain knowledge, data can be transformed into a powerful ally for antibiotic stewardship, potentially saving lives by eliminating the clinical "blind spot."